

Diego Atencia

AI Engineering — LLMs · RAG · AWS · Banking Data

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SUMMARY

M.Sc. in AI, B.Sc. in Mathematics (UAM). Experience in credit-risk (LGD/IRB) modeling and regulatory data pipelines (FINREP, COREP, CIRBE) in banking. Built and deployed regllm.xyz, a GenAI application for Spanish banking regulation (LoRA fine-tuning, RAG, AWS). Open-source contributions to LLM and RL infrastructure (Liger-Kernel, TorchRL).

EXPERIENCE

[1] Nov 2023 – Present
Madrid

- Built and validated LGD (Loss Given Default) credit-risk models for IRB portfolios, ensuring regulatory compliance.
- Developed Power Automate/Copilot Studio workflows integrating LGD model outputs into structured Word reporting pipelines.

SQL & PL/SQL Developer Oct 2022 – Nov 2023
Álamo Consulting — Madrid

- Owned regulatory reporting pipelines (FINREP, COREP, SIRBE/CIRBE) for an external banking entity using Oracle and T-SQL.
- Built scalable transformations within a large regulatory codebase, ensuring accurate and timely data delivery.

SELECTED PROJECTS

regllm.xyz — GenAI app for Spanish banking regulation Q&A Production, 2026

- Fine-tuned Qwen2.5-7B with LoRA; RAG pipeline; full-stack deployment on AWS (ECS Fargate, RDS/pgvector, ALB).
- Built end-to-end; serving real user queries in early access (~30 to date).

Hand Prosthesis Control with Deep RL — blog M.Sc. thesis

- Multimodal RL agent (PPO/DDPG) integrating stereoscopic vision and sensor data for real-time prosthetic hand control in MuJoCo simulation.

OPEN-SOURCE CONTRIBUTIONS

Liger-Kernel — github.com/linkedin/Liger-Kernel PR #1128 (under review)
Production CUDA kernel library for LLM training, used at scale across industry

- Debugging a distributed-training bug in LigerTiledSwiGLUMLP: FSDP's post-backward reshard hooks were triggered prematurely inside a tiling loop, corrupting parameter shards across GPUs.
- Adapting axolotl's gradient-accumulator approach as the fix.

TorchRL — github.com/pytorch/rl Discussion #3538
PyTorch RL library (Meta)

- Proposing and implementing MC-PILCO: a model-based RL algorithm using Gaussian Process world models and Monte Carlo rollouts.

LEANN Issue #41

- Proposed a hybrid BM25+RRF retrieval pipeline over the neural-only architecture to improve recall on sparse queries.

EDUCATION

M.Sc. in Artificial Intelligence (8.67/10) Sep 2024 – Oct 2025
International University of Valencia
Thesis: Optimizing low-cost hand prosthesis control using Deep Reinforcement Learning.

B.Sc. in Mathematics Sep 2018 – Jun 2023
Autonomous University of Madrid (UAM)

Final thesis: Weak solutions to Navier–Stokes equations (9/10). Founder of university AI student association (70+ members).

TECHNICAL SKILLS

ML / GenAI	PyTorch, TensorFlow, Scikit-learn, HuggingFace, LoRA/QLoRA, RAG pipelines
Languages	Python (expert), SQL (expert), C/C++ (strong)
Systems & Cloud	AWS, Docker, Linux, Git/GitHub, CI/CD principles, distributed training (FSDP/DDP), CUDA kernels
Automation	Power Automate, Copilot Studio
Databases	Oracle, PostgreSQL, ChromaDB (vector), SAS

LANGUAGES

Spanish (native) English (fluent, professional working proficiency) Mandarin Chinese (learning)